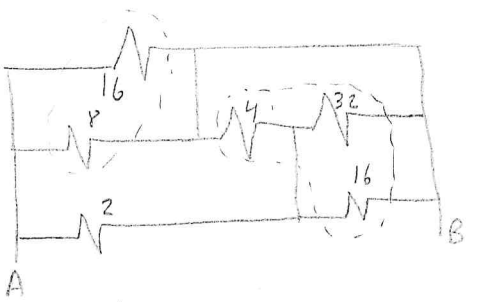
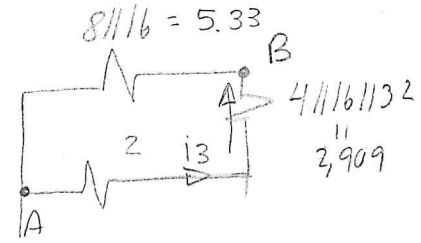
# PARCIAL OCTUBRE 2026 (GRUPO A)

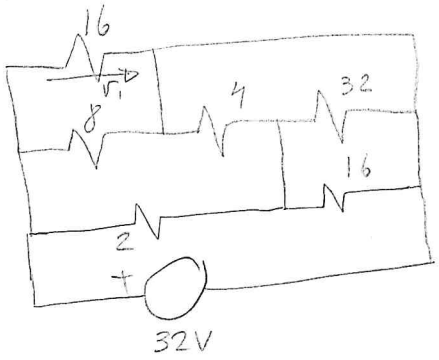
## 1




$R_{AB} = (2 + 2{,}909) // 5.33$

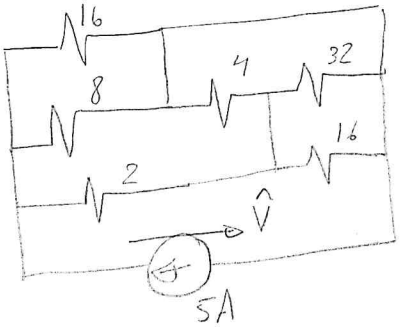
$R_{AB} = 2{,}55\,\Omega$




$i_3 = \frac{32}{2+2{,}909} = 6{,}518A$, $V_1 = 32V$

$V_4 = 32\ \frac{2{,}909}{2+2{,}909} = 18{,}96V$

$P_g = (32)^2 / 2{,}55 \Rightarrow P_g = 401{,}56W$



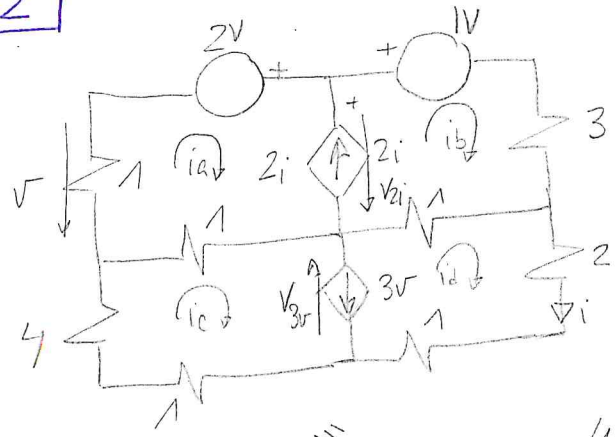

$\hat{V} = 5A\ 2{,}55\Omega = 12{,}75V$ $V_1 = 12{,}75\,V$

$V_4 = 12{,}75\ \frac{2{,}909}{2+2{,}909} = 7{,}55\,V$

$i_3 = \frac{12{,}75}{2+2{,}909} = 2{,}597A$

$P_g = 5^2 \cdot 2{,}55 = 63{,}75W$

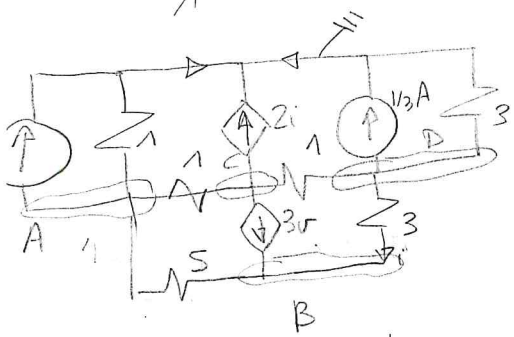
## 2




$$\begin{pmatrix} 2 & 0 & -1 & 0 \\ 0 & 4 & 0 & -1 \\ -1 & 0 & 6 & 0 \\ 0 & -1 & 0 & 4 \end{pmatrix} \begin{pmatrix} ia \\ ib \\ ic \\ id \end{pmatrix} = \begin{pmatrix} 2 - V_{2i} \\ -1 + V_{2i} \\ V_{3V} \\ -V_{3V} \end{pmatrix}$$

6 ec. 6 incog.

$ib - ia = 2\,id$ $\quad ic - id = -3\,ia$




$$\begin{pmatrix} 1+1+1/5 & -1/5 & -1 & 0 \\ -1/5 & 1/3+1/5 & 0 & -1/3 \\ -1 & 0 & 1+1 & -1 \\ 0 & -1/3 & -1 & 1+1/3+1/3 \end{pmatrix} \begin{pmatrix} V_A \\ V_B \\ V_C \\ V_D \end{pmatrix} = \begin{pmatrix} -2 \\ 3V \\ -3V - 2i \\ -1/3 \end{pmatrix}$$

$i = \frac{V_D - V_B}{3} \Rightarrow 2i = \frac{2}{3}(V_D - V_B)$

$2 + V = 0 - V_A \Rightarrow V = -2 - V_A$

$$\begin{pmatrix} 1+1+1/5 & -1/5 & -1 & 0 \\ 3\,1/5 & 1/3+1/5 & 0 & -1/3 \\ -3-1 & -2/3 & 1+1 & -1+2/3 \\ 0 & -1/3 & -1 & 1+1/3+1/3 \end{pmatrix} \begin{pmatrix} V_A \\ V_B \\ V_C \\ V_D \end{pmatrix} = \begin{pmatrix} -2 \\ -6 \\ 6 \\ -1/3 \end{pmatrix} \Rightarrow$$

$V_A = -1.958V$

$V_B = -2.063V$

$V_C = -1.896V$

$V_D = -1.75V$

$i = \frac{V_D - V_B}{3} = 0.104A$

$V = -2 - V_A = -0.042V$

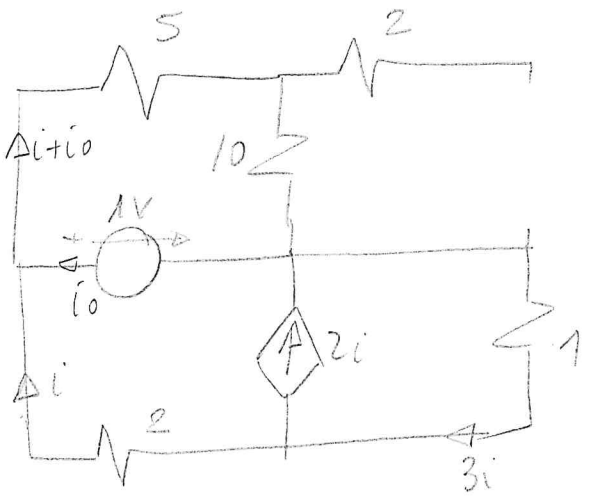
$P_{2V} = 2(2 + V_A) = 0.084W$ (gen)

$P_{1V} = 1 \cdot (1/3 + V_D/3) = -0.25W \rightarrow 0.25W$ (con)

$P_{2i} = 2i(0 - V_C) = 0.394W$ (gen)

$P_{3V} = 3V(V_B - V_C) = 0.021W$ (gen)

**3**




$1 = 15(i + io)$

$0 = 1 + 3i + 2i \Rightarrow i = -1/5$

$io = \frac{1}{15} - i \Rightarrow io = 0.2667$

$R_{th} = \frac{1}{io} = 3.75\Omega$

$R^{max} = 3.75\Omega$